

Machine Learning for Natural Language Understanding

Prof. Achim Rettinger

Lecturers

- Lecture: Achim Rettinger
- Tutorial: Simon Werner
- Seminar: Kai Kugler, Raghvi Baloni

- Our Research:

Data	Method	Application Areas
• Media Documents (multilingual text, multimodal) • Knowledge Graphs	• Machine Learning • Knowledge Representation	• Computational Social Sciences • Computational Communication Science

Logistics

Studienverlaufsplan

Modulkürzel	Modulbezeichnung (Pflicht/Wahlpflicht)	Sem.	LP	Typ	SWS	Veranstaltungsbezeichnung	Prüfung	Modul- beauftragte/r	Bemerkung/ Sprache
1. Semester (Wi)									
■	Preparation Course (WP)	Wi	20	Module im Umfang von 20 LP aus dem Bereich „Preparation Course“ sind zu absolvieren.					
■ Modulkürzel	Machine Learning for Natural Language Understanding (P)	Wi	10	V	2	Machine Learning for Natural Language Understanding		Rettinger	Englisch
				Ü	1	Machine Learning for Natural Language Understanding			
				S	2	Trends in Machine Learning			
				Schriftliche Ausarbeitung					

Who to contact

- General advice about Porta and enrolling in classes:
https://cl-trier.github.io/nlp_introduction_2025.pdf
- For any technical issues with Porta, please contact the Porta support (porta@uni-trier.de)
- For everything concerning official student documents and/or login credentials, please contact the admissions office (admission@uni-trier.de)
- For questions about the NLP study program: Simon Werner
- Information concerning this module (ML4NLU)
 - Lecture: Course on StudIP, Messages via StudIP, Achim Rettinger via office hours
 - Tutorial: Course on StudIP, Simon Werner
 - Seminar: Course on StudIP, Kai Kugler & Raghvi Baloni

Weekly logistics – 7 days, 1 topic

- Tuesday night:
 - New lecture video + questions are available on StudIP
 - Watch and prepare some answers for Thursday's lecture
- Thursday 10:15 (lecture, hybrid meeting, attendance voluntary):
 - Flipped classroom answers and discussions (max 30 min, only on-site)
 - Summary presentation (max 60 min, streamed via ZOOM)
- Thursday+Friday (seminar, on-site meeting, attendance compulsory)
 - group presentations + discussion by and for students
 - Papers related to lecture topic of the week
- Following Monday 10:15 (tutorial, hybrid meeting, attendance voluntary):
 - Exercises, Q&A, solutions
 - Practical implementation and running of algorithms, libraries and data
 - Questions, code and data related to lecture topic of the week

Next Sessions

- Lecture (Every Thursday 10:15-11:45 N2)
- The next session will be on Thursday October 23 via a hybrid setup. Limited personal attendance possible; unlimited online attendance; attendance not required.
<https://uni-trier.zoom-x.de/j/68859110758?pwd=Gf627nP0fo5v3lMGVqH6b5ViVcKC1v.1>
- Tutorials (Every Monday 10:15-11:45 N2)
- First tutorial sessions will be on October 20th. Hybrid setup; limited personal attendance possible; attendance not required.
- Seminars (Thursdays 12:15-13:45 and Fridays 08:15-09:45)
- We plan to have 2x2 parallel sessions. Weekly personal attendance and active participation is mandatory.

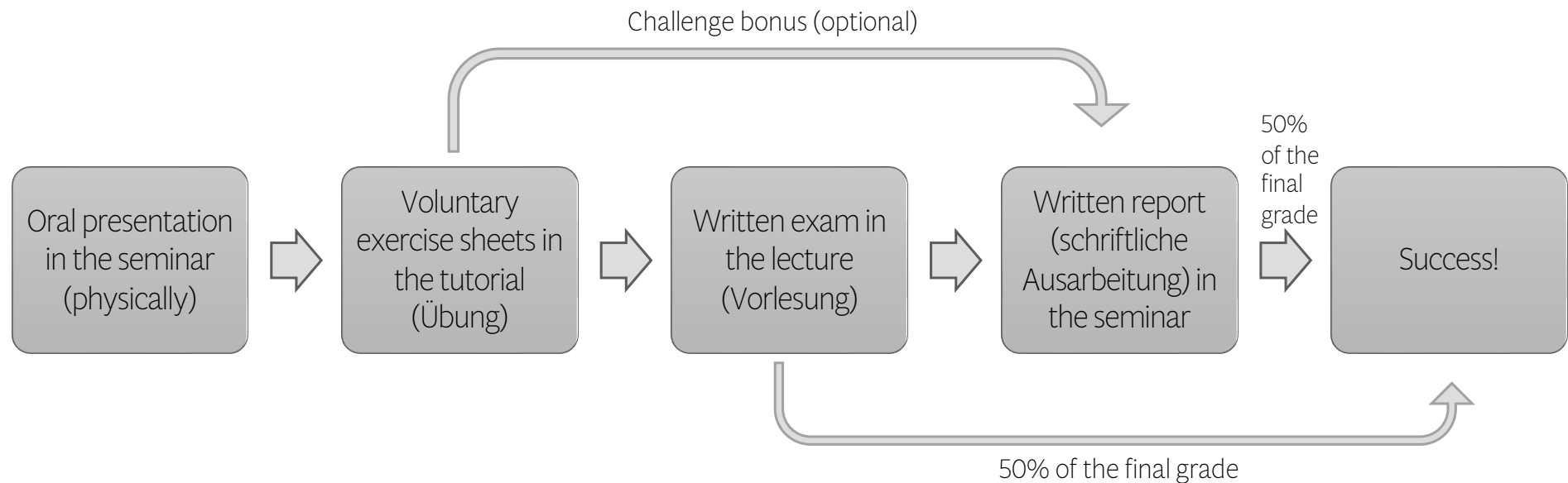
Lecture (Vorlesung)

- Semi-flipped classroom
 - First part (asynchronous and online): a list of online sources + questions will be published. Watch/read the content. Don't expect to understand everything and don't learn everything.
 - Second part (live and in person and on-site): Answer and discuss questions
 - Third part (live and hybrid (online and on-site): Summary presentation
- Test: written exam (counts 50% of the final grade)
- Main sources for lecture videos
 - Stanford CS224U: Natural Language Understanding
 - <https://youtube.com/playlist?list=PLoROMvodv4rOwvldxftJTmoR3kRcWkJBp>
 - Stanford CS224N: Natural Language Processing with Deep Learning
 - <https://youtube.com/playlist?list=PLoROMvodv4rMFqRtEuo6SGjY4XbRIVRd4>
 - Meta Llama tech reports
 - <https://ai.meta.com/research/publications/the-llama-3-herd-of-models/>
 - <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>

Detailed Logistics w.r.t. Grading

- Don't forget that you must sign up for all exams!
- 50% of the official „exam“ for the module is the „written report“. Due end of February. The officially listed deadline is the end of March (end of semester) in PORTA.
- The hard deadline for the “written report” is end of March. If you miss the hard deadline this automatically counts as a failed attempt, and you must sign up for a re-examination in the next semester.
- The written exam is a mixture of questions in the style of the “flipped classroom questions” and questions from the tutorial/exercise sheets (applied). It makes up 50% of the final grade.

Path to final grade for the ML4NLU module



Bonus policy

- Challenge bonus (will be subtracted from your written assignment grade)
 - Beating our baseline: one grading step
 - Handing in the best model: two grading steps

“The future of language is about the development of language models...”¹

1

Computerlinguistik | Universität Trier | cl.cldh.uni-trier.de